X-Ray Fluorescence Analysis

Parth Bhargava (A0310667E)
PC3193 Experimental Physics II

X-ray fluorescence (XRF) spectroscopy was used for qualitative elemental identification of unknown samples and quantitative composition analysis of a brass alloy. In Part 1, calibrated spectra from a Fe+Zn reference target were used to assign characteristic line energies to six unknowns: brass (Cu, Zn, Pb), a magnetic sample (Sm, Co), old/new 50-cent coins (Cu, Ni), an old 1-dollar coin (Cu-rich, no resolved Ni line), and a new 1-dollar coin (Cu, Ni). Observed line energies agreed with tabulated values to within ±0.08 keV in all cases, and K_β/K_α intensity ratios fell in the expected range of 1/5–1/10, confirming consistent identification. In Part 2, Gaussian deconvolution resolved the Cu K_β / Zn K_α overlap near 8.7 keV in the brass spectrum. Using the manual's first-order mass-ratio relation and five repeated trials, the brass composition was estimated as Cu (61 ± 8)%, Zn (37 ± 8)%, Pb (3 ± 3)%, consistent with leaded brass. The dominant source of uncertainty was run-to-run scatter in the Pb reference spectrum (H_0 ranging from 453 to 868), reflecting the sensitivity of the weak Pb L_α line to sample positioning.

1. INTRODUCTION

Knowing which elements are present in a material, and in what proportions, is fundamental to understanding its properties. The corrosion resistance of a coin, the machinability of a brass fitting, and the magnetic strength of a permanent magnet all depend on elemental composition. X-ray fluorescence (XRF) gives us this information without damaging the sample. The idea is straightforward: a primary X-ray knocks out a core electron from an atom, a higher-shell electron drops down to fill the hole, and the energy difference comes out as a characteristic X-ray photon whose energy is set by the atomic number. Since core electrons are tightly bound and barely influenced by chemical bonding, these emitted energies act as fingerprints for each element. A multi-element sample just gives a superposition of its individual elemental spectra, so by reading off the peak energies we can identify what is present and, with more work, how much of each element there is.

This experiment has two objectives:

- **Qualitative analysis (Part 1)**: identify the elements present in brass, a magnetic sample, and old/new Singapore coins by matching observed peaks to tabulated characteristic line energies and verifying self-consistency through K_β/K_α intensity ratios.
- **Quantitative analysis (Part 2)**: estimate mass fractions in brass from fitted Gaussian peak heights relative to pure-element reference spectra, using repeated measurements to assess reproducibility.

2. THEORY

2.1 Characteristic X-ray Emission

When an incident photon ejects a K-shell electron, the vacancy can be filled by an electron from the L-shell (K_α transition) or M-shell (K_β transition). For a given element, the K_β line is at higher energy than K_α because the M-to-K gap is larger than the L-to-K gap. The K_β line is also weaker, typically 1/5 to 1/10 of K_α, because the M→K transition is less probable.

For heavy elements like lead ($Z = 82$), the K-shell binding energy is 88 keV, far above the 35 kV tube voltage we use here. The tube simply cannot knock out a K-shell electron, so only L-series transitions are observed: L_α ($M \to L$, 10.55 keV), L_β ($N \to L$, 12.61 keV), and L_γ (14.76 keV).

Moseley's law provides the approximate scaling of line energy with atomic number:

$$\sqrt{E} = a(Z - \sigma) \tag{1}$$

where a and σ are constants for a given series. This monotonic relationship means that elements of similar Z produce lines close in energy. In particular, Ni K_β (8.265 keV) and Cu K_α (8.044 keV) are separated by only 0.22 keV and are unresolvable by a semiconductor detector with typical resolution $\Delta E \approx 0.2$ keV at these energies.

2.2 Quantitative First-Order Model

For a homogeneous sample of element i under X-ray irradiation, the measured peak height H_i should be proportional to how many atoms of that element sit in the irradiated volume. If the sample has cross-sectional area S and effective irradiation depth d, the number of atoms of element i is:

$$n_i = \frac{\rho_i S d}{A_i}$$

where ρ_i is the bulk density and A_i the atomic mass. If we ignore matrix effects for now (self-absorption, secondary fluorescence, and composition-dependent attenuation), the peak height is simply proportional to n_i:

$$H_i \propto n_i \propto \rho_i$$

For the same element measured as a pure reference, the peak height $H_{0,i}$ is proportional to its full bulk density ρ_i. So the ratio H_i / $H_{0,i}$ tells us roughly what fraction of the irradiated volume is occupied by element i. To turn this into a mass fraction we weight by the element's bulk density ρ_i, because denser elements contribute more mass per unit volume:

$$C_i = \frac{\rho_i H_i \,/\, H_{0,i}}{\sum_j \rho_j H_j \,/\, H_{0,j}} \tag{2}$$

For this to hold, every measurement needs the same geometry, and the detector response has to be roughly uniform across the energy range. In practice neither condition is perfectly met, and matrix effects (self-absorption and composition-dependent attenuation) push the results away from this simple first-order picture.

2.3 Gaussian Deconvolution of Overlapping Peaks

In the brass spectrum, Cu K_β ($\approx$ 8.90 keV) partially overlaps Zn K_α ($\approx$ 8.64 keV). If we just read peak heights straight from the spectrum, the Zn peak would be inflated by the Cu K_β tail sitting underneath it. The CASSY software deals with this by fitting the 7.5–9.1 keV region with multiple Gaussians:

$$N(E) = \sum_k A_k e^{-(E-\mu_k)^2 \,/\, (2\sigma^2)}$$

where A_k is the amplitude, μ_k the centre (fixed to the known line energy), and σ is constrained to be equal for all components within a fit region. This separates the Cu K_α, Cu K_β, and Zn K_α amplitudes for use in Eq. (2).

3. PART 1: QUALITATIVE ELEMENTAL IDENTIFICATION

3.1 Aim

Identify the elements present in six unknown samples (a brass alloy, a magnetic target, and four Singapore coins) by matching observed characteristic X-ray line energies to tabulated values and verifying each assignment through K_β/K_α intensity ratios or L-series line families.

3.2 Methods

The X-ray tube (Mo anode) was operated at 35 kV and 1 mA. Fluorescent X-rays were detected at 90° to the primary beam using a Si semiconductor detector coupled to a multi-channel analyser (MCA) with 512 channels. Each spectrum was recorded for 300 s. The target angle was set to 45° to maximise the irradiated area (see Table VI in Appendix).

Calibration was performed using the Fe+Zn reference target. The two strongest peaks, Fe K_α (6.403 keV) and Zn K_α (8.638 keV), were assigned in the CASSY Lab software to establish the linear channel-to-energy conversion. This two-point calibration defines the energy axis for all subsequent spectra.

After calibration, spectra were recorded for six samples by placing each in the sample holder and acquiring for 300 s:

1. Brass (Target 3)
2. Magnetic sample (Target 4)
3. Old 50-cent coin
4. New 50-cent coin
5. Old 1-dollar coin
6. New 1-dollar coin

Element identification used two criteria: (i) the observed peak energy was matched to tabulated values (Bearden 1967), and (ii) the presence of a consistent line family was verified: for K-series elements, both K_α and K_β should appear with the latter at $\approx$ 1/5–1/10 the intensity.

3.3 Results

The calibration spectrum (Fig. 1) shows the four expected peaks from the Fe+Zn target. Fe K_α and Zn K_α define the energy scale; Fe K_β and Zn K_β then serve as a check that the calibration is linear. The broad bump below $\approx$ 5 keV in all spectra is bremsstrahlung, the continuous X-ray background that comes from electrons decelerating in the Mo anode; the 35 kV tube voltage sets its upper cutoff. All the characteristic peaks we care about lie above 5.6 keV, where this background has mostly died away.

The calibration data also allow a check of the K_β/K_α intensity ratio. For Fe, this ratio is $333/1710 = 0.195 \approx 1/5.1$; for Zn it is $100/838 = 0.119 \approx 1/8.4$. Both sit in the expected 1/5–1/10 range, which gives us confidence that the line assignments are correct.

Fig. 2 shows the brass spectrum. Cu K_α and Zn K_α dominate, with partial overlap of Cu K_β and Zn K_α near 8.7 keV. A weak Pb L_α line is visible at 10.55 keV. Seeing Pb L_β (12.7 keV) and a marginal Pb L_γ (14.7 keV) as well confirms that the 10.55 keV peak really is Pb and not a coincidence.

The magnetic sample (Fig. 3) shows Co and Sm lines, which points to a SmCo-type rare-earth permanent magnet. The

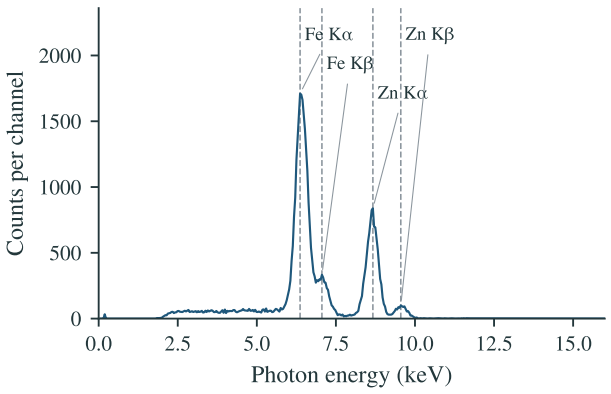


FIG. 1. Calibration spectrum (Fe + Zn target, 300 s), in counts per channel of the multichannel analyser. Dashed lines here and in Fig. 2 to Fig. 7 mark the measured line positions, labelled with their assignments. The Fe K_α (6.40 keV) and Zn K_α (8.67 keV) peaks define the linear energy calibration. Fe K_β and Zn K_β appear at approximately 1/5 and 1/8 the respective K_α intensities. The broad feature below $\approx$ 5 keV is the bremsstrahlung continuum from the Mo tube.

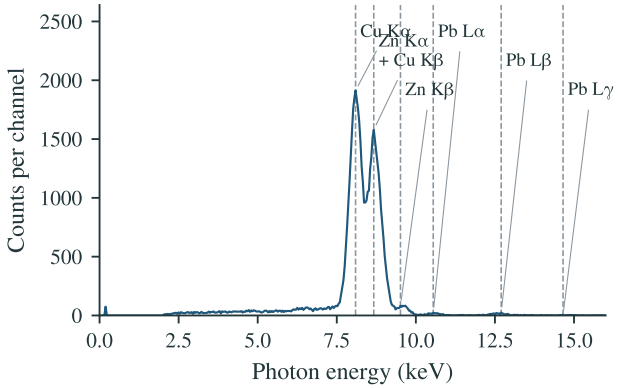


FIG. 2. Brass (Target 3) spectrum, 300 s. Cu K_α (8.09 keV) and Zn K_α + Cu K_β (merged near 8.7 keV) dominate. Pb is confirmed by the complete L-series: L_α (10.55 keV), L_β (12.7 keV), and L_γ (14.7 keV, barely above noise).

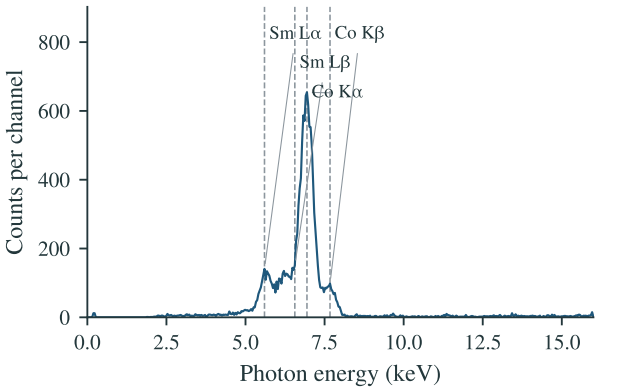


FIG. 3. Magnetic sample (Target 4) spectrum. The dominant feature is Co K_α (6.94 keV), with Co K_β at 7.67 keV. Sm is identified from its L_α (5.60 keV) and L_β (6.48 keV) lines.

Co K_β/K_α ratio of $98/654 = 0.150$ ($\approx 1/6.7$) sits right in the expected range.

The four coin spectra are shown in Fig. 4 through Fig. 6. The old and new 50-cent coins (Fig. 4 and Fig. 5) are both Cu-Ni alloys, with Ni K_α and Cu K_α clearly resolved. In these spectra, the Ni K_β (8.265 keV) and Cu K_α (8.044 keV) lines are separated by only 0.22 keV, which is below the detector resolution; they merge into a single unresolved peak labelled "Cu K_α + Ni K_β" in the plots.

The old 1-dollar coin (Fig. 7) shows only Cu lines, with no sign of Ni. The new 1-dollar coin (Fig. 6) shows both Ni and Cu.

Table I compares observed peak energies to literature values across the calibration and brass spectra. Deviations are within ± 0.08 keV for isolated lines. The two-point linear calibration introduces small nonlinear residuals at energies far from the calibration points, as seen in the $+0.08$ keV offset of Pb L_β at 12.7 keV.

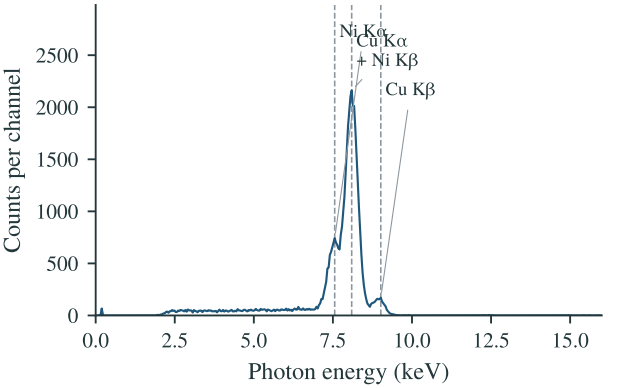


FIG. 4. Old 50-cent coin spectrum. The Ni K_α shoulder at 7.56 keV is clearly resolved from the dominant Cu K_α + Ni K_β blend at 8.09 keV, confirming a Cu-Ni alloy.

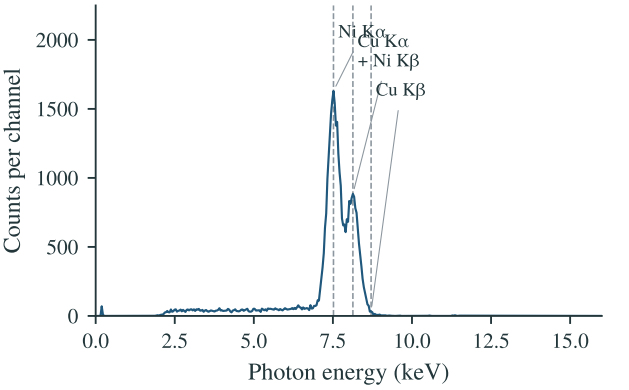


FIG. 5. New 50-cent coin spectrum. Similar Cu-Ni alloy signature as the old coin, with comparable relative peak intensities.

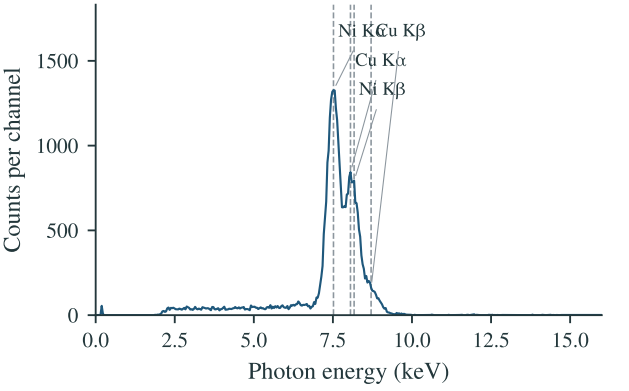


FIG. 6. New 1-dollar coin spectrum. Both Ni K_α (7.52 keV) and the Cu K_α / Ni K_β blend (8.1 keV) are visible, confirming a Cu-Ni alloy composition.

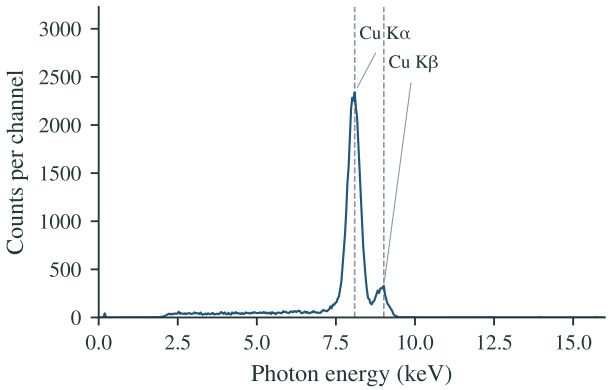


FIG. 7. Old 1-dollar coin spectrum. Only Cu K_α (8.09 keV) and Cu K_β (9.01 keV) are present. The Cu K_β/K_α ratio of $330/2338 = 0.141$ ($\approx 1/7.1$) is consistent with a pure Cu-based alloy.

TABLE I. Observed vs. literature (Bearden 1967) line energies. The calibration assigns the literature energies of Fe K_α and Zn K_α to specific MCA channels; "observed" values are the energies of the peak-maximum bin, which may differ by up to one bin width (≈ 0.03 keV). Deviations in other lines reflect residual nonlinearity of the channel-to-energy mapping.

Line	Literature (keV)	Observed (keV)	Δ (keV)
Fe K_α	6.403	6.367	−0.036
Fe K_β	7.057	7.058	+0.001
Zn K_α	8.638	8.669	+0.031
Zn K_β	9.572	9.551	−0.021
Cu K_α (brass)	8.044	8.094	+0.050
Pb L_α (brass)	10.551	10.549	−0.002
Pb L_β (brass)	12.614	12.697	+0.083

TABLE II. Qualitative identification summary. Energies are observed peak centres; assignment uses line-energy matching and line-family consistency.

Sample	Main observed lines (keV)	No. elements	Identified composition
Brass (Target 3)	8.09, 8.67, 10.55, 12.70	3	Cu + Zn + Pb
Magnetic sample (Target 4)	5.60, 6.48, 6.94, 7.67	2	Sm + Co
Old 50 cents	7.56, 8.09, 9.01	2	Ni + Cu
New 50 cents	7.52, 8.13, 8.75	2	Ni + Cu
Old 1 dollar	8.09, 9.01	1	Cu-rich alloy (no resolved Ni)
New 1 dollar	7.52, 8.06, 8.75	2	Ni + Cu

3.4 Analysis

3.4.1 Calibration Quality

The two-point calibration assumes a linear channel-to-energy conversion. The calibration assigns Fe K_α (6.403 keV) and Zn K_α (8.638 keV) to specific MCA channels, but when the peak-finding algorithm reads back the highest-count bin, the result can differ from the assigned energy by up to one channel width (≈ 0.03 keV), which is why even the anchor lines show small offsets in Table I. For non-anchor lines near the calibration range (6–10 keV), deviations reach at most ±0.05 keV. At higher energies (Pb L_β at 12.7 keV), the deviation grows to +0.08 keV, consistent with slight detector nonlinearity at the edge of the calibrated range. These shifts are small compared to inter-element line spacings (> 0.5 keV between neighbouring elements), so they do not compromise qualitative identification.

3.4.2 Detector Resolution

The Fe K_α peak in the calibration spectrum is well-isolated from its neighbours, making it a clean probe of the detector's energy resolution. From the Gaussian fit, the full width at half maximum (FWHM) is approximately 200 eV (0.20 keV) at 6.4 keV, which falls squarely within the 150–250 eV range typical for Si-PIN detectors at these energies. This number sets a hard limit on what we can resolve: any two lines closer than about 0.2 keV will blend together, which is exactly the situation with Ni K_β (8.265 keV) and Cu K_α (8.044 keV), separated by only 0.22 keV. It also explains why the Gaussian deconvolution in Part 2 is necessary for the Cu K_β / Zn K_α overlap (0.26 keV separation), where the peaks are distinguishable but not fully resolved.

3.4.3 Element Assignments

The brass spectrum contains three elements: Cu (K_α, K_β), Zn (K_α), and Pb (complete L-series: $L_\alpha, L_\beta, L_\gamma$). For Pb, seeing the full L-series matters because the individual lines are weak and could be mistaken for background features; three lines from the same series rules out a coincidence. It is also worth noting that no Pb K-lines appear anywhere in the spectrum. This is not because we missed them: the K-shell binding energy of Pb is 88 keV, far above the 35 kV tube voltage, so the tube physically cannot eject a K-shell electron from Pb. Only L-shell vacancies can be created at this voltage, and that is exactly what we observe.

The magnetic sample shows Co (with $K_\beta/K_\alpha = 0.150 \approx 1/6.7$, right in the expected range) and Sm (L_α at 5.60 keV, L_β at 6.48 keV), consistent with a SmCo-type rare-earth permanent magnet.

3.4.4 Unresolved Overlaps in Coin Spectra

In the Cu-Ni coin spectra, Ni K_β (8.265 keV) and Cu K_α (8.044 keV) are separated by only 0.22 keV. With the Si detector's resolution of ≈ 0.2 keV FWHM, these lines cannot be individually resolved and the merged peak is a blend of both transitions. We can still tell the two elements apart because each has another line that *is* resolved: Ni by its K_α at 7.48 keV and Cu by its K_β at 8.9 keV. Getting a quantitative Cu-to-Ni ratio for the coins would require Gaussian deconvolution like what was done for brass.

3.4.5 Coin Metallurgy

The old and new 50-cent coins both show Cu + Ni, consistent with cupronickel alloy (typically 75% Cu, 25% Ni). The old 1-dollar coin shows Cu only, consistent with the aluminium bronze composition (Cu-Al-Zn) used in pre-2013 Singapore dollar coins; Al (K_α at 1.49 keV) falls below the detector's practical sensitivity range. The new 1-dollar coin shows Cu + Ni, indicating a switch to a nickel-plated steel or cupronickel composition.

XRF probes only the near-surface layer (penetration depth ≈ 10–100 μm for metals at these energies). The new 50-cent and new 1-dollar coins are ferromagnetic, whereas the old coins are not. The old coins (cupronickel and aluminium bronze) contain no ferromagnetic constituent, so they are not attracted to a magnet. The new coins carry a steel core beneath a Cu-Ni cladding, and steel is ferromagnetic. We can quantify why no Fe signal appears: the mass attenuation coefficient of Cu at the Fe K_α energy (6.4 keV) is about 290 cm^2/g (NIST XCOM), giving an attenuation length of $1/(290 \times 8.96) \approx 3.8$ μm in copper. Through even a 20 μm Cu-Ni cladding, the Fe K_α intensity is reduced by a factor of $e^{20/3.8} \approx 190$, leaving less than 1% of the original signal. So any Fe characteristic X-rays from the steel core are almost completely absorbed by the cladding before reaching the detector. The fact that we see no Fe lines does not rule out a steel core; it is exactly what we would expect, and the magnetism of the new coins confirms the steel is there.

3.5 Discussion

Qualitative XRF identification is quick and non-destructive: 300 s per sample, no preparation needed. Even with a modest-resolution Si detector, checking both line energies and line-family consistency is enough for reliable identification, since every element in this experiment showed at least two lines from the same series.

The main limitations come down to detector resolution and energy range. The ≈ 0.2 keV FWHM resolution means that elements whose lines sit closer than this (like Ni K_β and Cu K_α, only 0.22 keV apart) blend into one peak, and you cannot do quantitative analysis without deconvolution. At the low-energy end, K-series lines below ≈ 3 keV get absorbed by the detector window and are lost under the bremsstrahlung background, so in practice we cannot see elements lighter than roughly potassium ($Z \geq 19$, $K_\alpha = 3.31$ keV). Al, for instance, has K_α at 1.49 keV and is completely invisible here, which is why the old 1-dollar coin appears to contain only Cu even though aluminium bronze is a Cu-Al-Zn alloy.

3.6 Improvements

- Use a silicon drift detector (SDD) for better energy resolution (≈ 0.13 keV FWHM at 5.9 keV vs. ≈ 0.2 keV for Si-PIN), which would resolve the Ni K_β / Cu K_α overlap.
- Add more calibration points (e.g. a multi-element standard) to improve linearity across a wider energy range, reducing the +0.08 keV residual at 12.7 keV.
- Include a low-energy window or He-flush path to extend sensitivity toward lighter elements (Al, Si, S) that are invisible with the current setup.

4. PART 2: QUANTITATIVE BRASS COMPOSITION

4.1 Aim

Estimate the mass fractions of Cu, Zn, and Pb in the brass alloy using Gaussian deconvolution and the first-order mass-ratio model (Eq. (2)), and assess the reproducibility of the method through repeated measurements.

4.2 Methods

The CASSY software's Gaussian fitting tool was used to decompose the overlapping Cu–Zn region of the brass spectrum. The fit region (channels ≈ 195–240) was modelled with three Gaussians (Cu K_α, Zn K_α, Cu K_β) sharing a common width, with centres fixed at the known line energies. A separate single-Gaussian fit was used for Pb L_α (channels ≈ 260–285).

This procedure was repeated five times (trials 2–6) for the brass sample and four times each for the pure Cu, Zn, and Pb reference targets, repositioning the sample between trials to capture the effect of geometric variability. All fit parameters were saved as CASSY `.labx` project files, from which the Gaussian amplitudes (A_k) were extracted programmatically.

4.3 Results

Fig. 8 shows the Gaussian deconvolution of the brass spectrum. The fit pulls apart the Cu K_α, Zn K_α, and Cu K_β contributions in the overlap region; the sum of the individual Gaussians (dashed) tracks the measured spectrum (solid) closely.

Table III lists the extracted Gaussian amplitudes from each trial, showing the spread that propagates into the final composition uncertainty.

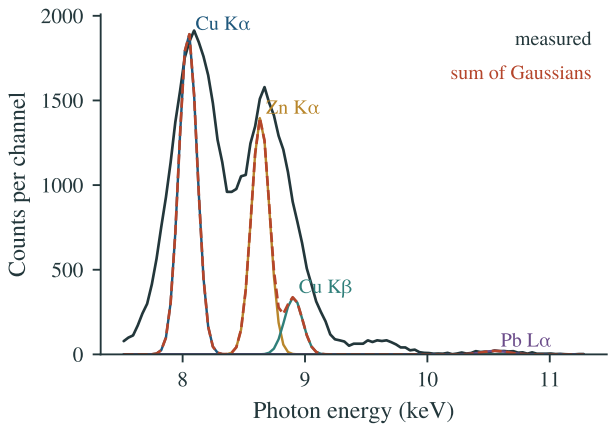


FIG. 8. Gaussian deconvolution of the brass spectrum in the 7.5–11.3 keV region. The Cu K_α (blue), Zn K_α (gold), Cu K_β (teal), and Pb L_α (purple) Gaussians are shown individually, with their sum (dashed rust) overlaid on the measured spectrum (black). Gaussian amplitudes have been scaled to match the Part 1 qualitative spectrum.

TABLE III. Gaussian fit amplitudes (H_i) for brass across five trials. The $\approx 1.6\times$ variation in Cu and Zn amplitudes reflects repositioning between trials.

Trial	H_{Cu}	H_{Zn}	H_{Pb}
Brass trial 2	2742.6	2005.3	29.5
Brass trial 3	1979.8	1474.7	23.3
Brass trial 4	1797.2	1253.8	21.9
Brass trial 5	2520.2	1906.9	23.5
Brass trial 6	1744.1	1243.3	19.4
Mean ± s.d.	**2156.8 ± 448.9**	**1576.8 ± 360.0**	**23.5 ± 3.7**

TABLE IV. Reference spectrum amplitudes ($H_{0,i}$) for pure elements. Pb shows the widest spread (452–868), reflecting geometric sensitivity of the weak L_α line.

Element	**Trial 1**	**Trial 2**	**Trial 3**	**Trial 4**
Cu (H_0)	1976.5	1894.9	1760.7	1688.3
Zn (H_0)	1714.2	1606.9	1860.6	1891.4
Pb (H_0)	867.5	590.1	481.2	452.8

Using the mean values from Table III and Table IV in Eq. (2) yields the mass fractions in Table V.

TABLE V. Mass-ratio calculation for brass using Eq. (2) and mean fitted heights from Table III and Table IV.

Element	ρ **(g/cm³)**	**Line**	$H\,/\,H_0$	$\rho \cdot H\,/\,H_0$	C_i **(%)**
Cu	8.96	K_α	1.179	10.56	60.8
Zn	7.14	K_α	0.892	6.37	36.6
Pb	11.34	L_α	0.039	0.45	2.6

Uncertainties were estimated by Monte Carlo propagation. For each of 50 000 iterations, the brass amplitudes (H_{Cu}, H_{Zn}, H_{Pb}) and the reference amplitudes ($H_{0,\mathrm{Cu}}$, $H_{0,\mathrm{Zn}}$, $H_{0,\mathrm{Pb}}$) were each drawn independently from normal distributions whose means and standard deviations were taken from the trial data in Table III and Table IV. Each draw was fed through Eq. (2) to produce one set of mass fractions; the reported uncertainties are the standard deviations of the resulting distributions:

- Cu: $(60 \pm 8)\%$
- Zn: $(37 \pm 8)\%$
- Pb: $(3 \pm 3)\%$

Most of the uncertainty comes from the scatter in the reference amplitudes (H_0), especially for Pb where H_0 ranges nearly $2\times$ across four trials. With only 4–5 trials per element, the normal approximation is rough; the true run-to-run distribution could easily be skewed or heavier-tailed, which would widen the confidence intervals further.

4.4 Analysis

The result (Cu ≈ 61%, Zn ≈ 37%, Pb ≈ 2–3%) makes sense for leaded brass: standard C36000 free-cutting brass is nominally 61.5% Cu, 35.5% Zn, 3% Pb, so our numbers are in the right ballpark.

The Cu H/H_0 ratio of 1.179 is above 1.0, which should not happen since the alloy has less Cu per unit volume than the pure reference. What most likely happened is that the sample was slightly differently positioned between the brass and reference runs, changing the geometric factor (solid angle × irradiated area). The normalisation in Eq. (2) saves us here: dividing each element's $\rho \cdot H/H_0$ by the sum over all elements makes the geometric factor cancel to first order, so the final mass fractions still come out reasonable.

The Cu:Zn ratio stays stable across trials ($H_{\mathrm{Cu}}/H_{\mathrm{Zn}} = 1.32$ –1.43) even though absolute amplitudes swing by a factor of $1.6\times$, showing that geometric effects largely cancel in relative measurements. This is why the ratio-based approach gives reasonable results despite the large absolute scatter.

The Gaussian fit in Fig. 8 shows a systematic shortfall in the 8.5–9.0 keV overlap region: the sum of the individual Gaussians (dashed) undershoots the measured spectrum (solid) by roughly 10% at the Zn K_α + Cu K_β peak. This probably happens because the three-Gaussian model, with its shared width and fixed centres, is too simple to capture the slightly asymmetric peak shapes and the channel-binning of the real detector. If the fit underestimates the Zn K_α amplitude, the resulting Cu:Zn ratio would be biased high. The CASSY software does not report a chi-squared statistic, so the fit quality can only be judged visually; the $\approx 10\%$ mismatch is an additional systematic uncertainty not captured by the Monte Carlo propagation.

Pb is the weakest part of the analysis. The Pb L_α peak has only ≈ 23 counts above background per trial, so Poisson noise alone is already $\sigma/N \approx 21\%$. On top of that, the Pb reference amplitude H_0 swings nearly $2\times$ across four trials (453–868), far more than Poisson statistics can explain ($\sigma/N \approx 2$–5% for peaks that big). The geometric sensitivity is amplified because the L_α line is so weak to begin with. The Monte Carlo uncertainty on Pb (±3.3% on a 3.1% value) means we can barely claim the Pb content is significant at all.

4.5 Discussion

Matrix effects. Eq. (2) ignores several things that happen inside a real sample. Self-absorption is the most obvious: a Cu K_α photon (8.04 keV) on its way out of the sample can be absorbed by another Cu atom before reaching the detector. In the alloy, where Cu is diluted, this happens less than in the pure reference where the sample is 100% Cu, so self-absorption distorts the H/H_0 ratio. There is also secondary fluorescence to think about: a characteristic photon from one element can excite fluorescence in another if the photon energy exceeds that element's absorption edge. In brass, Cu K_α (8.04 keV) is below the Zn K-edge (9.66 keV), so Cu does not enhance Zn. The reverse pathway does exist, though: Zn K_β (9.57 keV) sits just above the Cu K-edge (8.98 keV), so Zn fluorescence within the sample can knock out additional Cu K-shell electrons. In the brass alloy this means H_{Cu} picks up extra counts that are not present when measuring the pure Cu reference (which contains no Zn), and this could partly explain why the measured H/H_0 for Cu exceeds 1.0. In alloys containing Ni (K-edge 8.33 keV), Zn K_α (8.64 keV) could similarly excite secondary Ni fluorescence. The Si detector efficiency also varies with photon energy, though this partially cancels in the ratio $H_i/H_{0,i}$ since alloy and reference are measured with the same detector. These effects are small enough for a first-order estimate but can shift reported compositions by several percent.

Geometric variability vs. counting statistics. Photon counting follows Poisson statistics, so $\sigma_N = \sqrt{N}$. For a strong peak like Cu K_α ($N \approx 2000$), that gives $\sigma/N \approx 2.2\%$, far smaller than the $\approx 20\%$ scatter we see from repositioning the sample between runs. For the weak Pb L_α ($N \approx 23$), counting noise alone is already $\sigma/N \approx 21\%$. Background subtraction makes things worse:

$$\sigma^2_{\text{net}} = \sigma^2_{\text{peak}} + \sigma^2_{\text{bkg}}$$

so a weak line sitting on a comparable background gets hit twice. Signal counts grow linearly with acquisition time t but SNR only improves as $\sqrt{t}$:

- **Halving to 150 s** would reduce Cu K_α counts from ≈ 1900 to ≈ 950, raising its Poisson noise from 2.3% to 3.2%, which is still acceptable for strong lines. However, Pb L_α counts would fall to ≈ 12, giving $\sigma/N \approx 29\%$, making the Pb determination unreliable.
- **Doubling to 600 s** would improve the Pb L_α Poisson contribution from $\approx 21\%$ to $\approx 15\%$ ($\sqrt{46}/46$). The Cu:Zn ratio would be effectively unchanged since its uncertainty is dominated by geometric scatter, not counting statistics.

So for improving reproducibility, fixing the sample geometry matters more than counting for longer. That said, longer acquisitions do specifically help Pb, where Poisson noise is a real problem on its own.

4.6 Improvements

- Use a fixed sample-positioning jig to eliminate geometric variability between trials, addressing the dominant source of uncertainty.
- Increase counting time to 600 s or longer, particularly for the Pb reference, to improve Poisson statistics on weak L-series lines.
- Apply a matrix correction method (e.g. fundamental parameters or empirical alpha coefficients) instead of the first-order peak-height model, to account for self-absorption and secondary fluorescence.
- Use Gaussian deconvolution on the coin spectra to extract quantitative Ni:Cu ratios, enabling direct comparison between old and new coin compositions.

5. CONCLUSION

We identified all constituent elements across six unknowns using characteristic line energies and K_β/K_α ratio consistency checks, with observed energies agreeing with tabulated values to within ± 0.08 keV. The coin spectra also showed a practical limit: when two elements have lines within the detector resolution (Ni K_β and Cu K_α, 0.22 keV apart), we can still tell them apart through other lines in the family, but getting a quantitative ratio requires deconvolution.

For brass, Gaussian deconvolution of the Cu K_β / Zn K_α overlap gave Cu $\approx 61\%$, Zn $\approx 37\%$, Pb $\approx$ 2–3%, which matches standard leaded brass (C36000: 61.5/35.5/3%). The main source of uncertainty turned out to be run-to-run geometric variability rather than counting statistics: absolute peak heights swung by up to 1.6 × across trials. But the Cu:Zn ratio stayed stable (1.32–1.43), which shows that the ratio-based normalisation of Eq. (2) is much more robust than looking at absolute intensities. The Pb result, with a relative uncertainty exceeding 100%, makes clear that first-order XRF cannot reliably quantify trace constituents when both the signal and reference are this weak.

Overall, XRF is quick and non-destructive, good at identifying major constituents and giving reasonable composition estimates. But for trace-level accuracy or resolving closely spaced lines, you need a better detector or a more involved analysis method.

APPENDIX

Instrument Settings

TABLE VI. Operating settings used for XRF measurements.

Parameter	Value
Tube anode	Mo
Tube voltage	35 kV
Tube current	1 mA
Measurement channels	512
Counting time	300 s
Detector	Si semiconductor
Detector angle	90°
Target angle	45°
Calibration target	Fe + Zn

REFERENCES

[1] X-Ray Fluorescence Analysis, (n.d.).
[2] J. A. Bearden, X-ray wavelengths, Rev. Mod. Phys. **39**, 78 (1967).
[3] R. Jenkins, R. W. Gould, and D. Gedcke, *Quantitative X-Ray Spectrometry*, 2nd ed. (Marcel Dekker, 1995).
[4] R. Van Grieken and A. Markowicz, editors, *Handbook of X-Ray Spectrometry*, 2nd ed. (Marcel Dekker, 2002).
[5] B. D. Cullity and S. R. Stock, *Elements of X-Ray Diffraction*, 3rd ed. (Prentice Hall, 2001).

DECLARATION ON THE USE OF GENERATIVE AI

I declare that I **HAVE** used generative AI tools to produce this assignment.

I acknowledge that generative AI was used in the following manner:

TABLE VII. AI Tool Usage Declaration

AI Tool Used	**My Prompt and AI Output**	**How the Output Was Used**
Cursor / Claude Code	Prompt: "Process XRF CSV/LABX files, extract Gaussian fit parameters, and generate publication-quality plots and summary tables." Output: Python script and processed numerical summaries.	Used for data extraction/plotting automation. Final interpretation and report writing were verified and edited by me.
Cursor / Claude Code	Prompt: "Help structure the Typst report and format equations, figures, and tables for XRF analysis." Output: Typst layout guidance and draft text structure.	Used as drafting assistance. Physical interpretation, critical discussion, and final conclusions were written and checked by me.